Towards a First-Principles Derivation of the S-Curve Model for Quantum Error-Correction Logical Error Rates

Formalised in Lean 4 + mathlib4

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Abstract

We investigate whether the empirical three-parameter "S-curve" fit of the weight-conditional logical error rate P_L^w for stabilizer QEC codes under perfect minimum-weight decoding can be derived from first principles. In a Lean 4 formalisation built on mathlib, we set up a minimal abstract framework (stabilizer codes, perfect MWPM decoders, P_L^w as a rational) and prove four structural properties unconditionally: $P_L^w = 0$ below threshold, $P_L^w \in [0, 1]$, vanishing on empty failure sets, and the trivial monotonicity below the fault-tolerance threshold. We then *reduce* the general monotonicity question to a purely integer combinatorial inequality (the pair-cardinality inequality), and prove the reduction formally. Full monotonicity for an abstract stabilizer code is known to be false (small-code counterexamples in the data-only setting); our target setting is the circuit-level compiled surface code, where the pair-cardinality inequality is expected to hold by empirical evidence but remains a research-level open problem. The Lean library is small, self-contained, and compiles with a single documented `sorry` on the open combinatorial step.

1 Introduction

The weight-conditional logical error rate P_L^w — the probability that a uniformly random weight-w fault configuration causes the decoder to output a logical error — appears empirically to follow a three-parameter S-shaped curve in w [1]. Prior work fits this S-curve ansatz to Monte-Carlo data and uses it for logical-error-rate extrapolation. Here we ask: *can the S-curve form be derived rigorously?*

This paper reports the current state of that derivation. The answer so far is *partial*: the structural preliminaries are cleanly formalisable, the monotonicity question reduces cleanly to a single integer inequality, and the remaining combinatorial step is concrete but research-level.

Contributions.

1. A minimal abstract formalism (Lean 4 + mathlib) for stabilizer codes, perfect MWPM decoders, and the weight-conditional LER P_L^w, based only on the definition of code distance.
2. Unconditional formal proofs of:
   - $P_L^w = 0$ for $w \leq t$ (fault-tolerance below threshold);
   - $0 \leq P_L^w \leq 1$ (range);
   - monotonicity below threshold;
   - a clean saturation lemma ($P_L^w = 0$ when the failure set is empty).

3. A formal *reduction* of the step-monotonicity claim $P_L^w \le P_L^{w+1}$ to the integer inequality

$$|\mathrm{fail}(w)| \cdot |\mathrm{wt}(w+1)| \;\le\; |\mathrm{fail}(w+1)| \cdot |\mathrm{wt}(w)|.$$

4. An honest report on the open questions: general monotonicity fails in the abstract stabilizer-code setting (with small-code counterexamples discovered by brute-force search), and the full S-curve approximation theorem is left for future work.

Companion Lean library. All proofs are formalised in the Lean 4 library available at `https://github.com/yezhuoyang/SCurveQEC-Lean`. Each named theorem in this paper cross-references the corresponding Lean identifier. The library contains one documented `sorry`, isolated to the code-specific combinatorial step described in Section **??**.

2 Preliminaries

We work at the level of an abstract stabilizer code and a fault-tolerant decoder; no concrete code family (surface, toric, ...) is formalised at this stage.

Definition 2.1 (Pauli operator)**.** For $n \in \mathbb{N}$, an n-qubit Pauli is a function $P : \{0, \dots, n-1\} \to \{I, X, Y, Z\}$. We write $\mathrm{wt}(P)$ for the number of indices where $P(i) \neq I$, and multiplication is pointwise (modulo global phase, which we suppress throughout since logical error rates are phase-independent).

Lean: `SCurveQEC.SinglePauli`, `SCurveQEC.Pauli`, `Pauli.weight`.

Definition 2.2 (Stabilizer code)**.** A stabilizer code on n qubits is a 4-tuple $\mathcal{C} = (n, \mathcal{S}, \mathcal{L}, \mathrm{axioms})$ where $\mathcal{S} \subseteq \mathcal{P}_n$ is a set of Paulis containing the identity and $\mathcal{L}$ is a superset of $\mathcal{S}$ that contains the identity and at least one non- stabilizer element. Elements of $\mathcal{L} \setminus \mathcal{S}$ are the *logical-error operators*. The *distance* of $\mathcal{C}$ is

$$d(\mathcal{C}) := \inf\{\, \mathrm{wt}(L) : L \in \mathcal{L} \setminus \mathcal{S} \,\},$$

and the *fault-tolerance threshold* is $t(\mathcal{C}) := \lfloor (d-1)/2 \rfloor$.

Lean: `StabilizerCode` (structure), `.distance`, `.threshold`.

Definition 2.3 (Perfect MWPM decoder)**.** A perfect minimum-weight decoder D for $\mathcal{C}$ is a function $D : \mathcal{P}_n \to \mathcal{P}_n$ satisfying:

- $D(E) \cdot E \in \mathcal{S} \cup \mathcal{L}$ (coset consistency);
- $\mathrm{wt}(D(E)) \le \mathrm{wt}(F)$ for every F with $F \cdot E^{-1} \in \mathcal{S}$ (minimum weight);
- $D(E_1) = D(E_2)$ whenever $E_1 \cdot E_2^{-1} \in \mathcal{S}$ (syndrome determinism).

Lean: `PerfectMWPM` (structure, three axioms).

Definition 2.4 (Weight-conditional logical error rate)**.** For $w \in \{0, 1, \dots, n\}$,

$$P_L^w(\mathcal{C}, D) := \frac{|\{E : \mathrm{wt}(E) = w \text{ and } D(E) \cdot E \in \mathcal{L} \setminus \mathcal{S}\}|}{|\{E : \mathrm{wt}(E) = w\}|},$$

with the convention $P_L^w = 0$ if the denominator is zero.

Lean: `logicalFailures`, `P_L`; basic inequalities proved as `P_L_nonneg`, `P_L_le_one`.

3 Structural Properties of P_L^w

We establish four structural results on P_L^w. Each is proved unconditionally in Lean.

3.1 Fault-Tolerance Below Threshold

Theorem 3.1 (Fault-tolerance below threshold; Lean: `Thm_1A_fault_tolerance`)**.** *For any stabilizer code $\mathcal{C}$ and perfect MWPM decoder D, if $w \le t(\mathcal{C})$ then $P_L^w(\mathcal{C}, D) = 0$.*

Proof. It suffices to show that the decoder succeeds on every weight-w Pauli with $w \le t$. Let E be such an error and set $F := D(E)$. *Step 1.* $\mathrm{wt}(F) \le \mathrm{wt}(E)$, by the min-weight axiom applied with $F' := E$ (noting that $E \cdot E^{-1} = 1 \in \mathcal{S}$). *Step 2.* $\mathrm{wt}(F \cdot E) \le 2\mathrm{wt}(E) \le 2t$, by the triangle inequality for Pauli weight (Lean: `Pauli.weight_mul_le`). *Step 3.* $F \cdot E \in \mathcal{S} \cup \mathcal{L}$, by the coset-consistency axiom. *Step 4.* Since $d \ge 2t + 1$, every non-stabilizer logical has weight at least $d > 2t$; Steps 2 and 3 then force $F \cdot E \in \mathcal{S}$ (Lean: `stab_of_small_weight_logical`). □ □

3.2 Range

Theorem 3.2 (Range; Lean: `Thm_1B_range`)**.** *For any $\mathcal{C}, D$ and any w, $0 \le P_L^w(\mathcal{C}, D) \le 1$.*

Proof. P_L^w is a ratio of Finset cardinalities. Non-negativity is immediate. The upper bound follows because the Finset of weight-w logical failures is a subset of all weight-w Paulis. (Lean: `P_L_nonneg`, `P_L_le_one`.) □ □

3.3 Monotonicity Below Threshold

Theorem 3.3 (Trivial monotonicity below threshold; Lean: `Thm_1C_monotonicity_below_threshold`)**.** *If $w_1 \le t$ and $w_1 \le w_2$, then $P_L^{w_1} \le P_L^{w_2}$.*

Proof. By Theorem 3.1, $P_L^{w_1} = 0$, and by Theorem 3.2 $P_L^{w_2} \ge 0$. □ □

3.4 Zero-on-Empty and Range Summary

Theorem 3.4 (Zero when failure set is empty; Lean: `Thm_1D_zero_when_no_failures`)**.** *If at weight w there are no logical failures, then $P_L^w(\mathcal{C}, D) = 0$.*

Proof. Immediate from the definition: the numerator's Finset is empty, so its cardinality is zero. □ □

4 Monotonicity: Reduction to a Combinatorial Inequality

The general monotonicity question, "does $P_L^{w_1} \le P_L^{w_2}$ for all $w_1 \le w_2$?", does not have a universally positive answer in the abstract stabilizer-code setting; we discuss this in Section **??**. In this section we give a clean *reduction*: monotonicity reduces to an integer inequality on Finset cardinalities, which can be verified per code.

4.1 The Pair-Cardinality Inequality

Definition 4.1 (Pair-cardinality inequality; Lean: `PairCardIneq`)**.** For a stabilizer code $\mathcal{C}$, perfect MWPM decoder D, and weight w,

$$\mathrm{PairCardIneq}_{\mathcal{C},D}(w) \quad \Longleftrightarrow \quad |\mathrm{fail}(w)| \cdot |\mathrm{wt}(w+1)| \;\leq\; |\mathrm{fail}(w+1)| \cdot |\mathrm{wt}(w)|,$$

where $\mathrm{fail}(w)$ is the Finset of weight-w Paulis producing a logical error under D, and $\mathrm{wt}(w)$ is the Finset of all weight-w Paulis.

4.2 Reduction Theorems

Lemma 4.2 (Arithmetic reduction; Lean: `monotone_step_via_cards`)**.** *Let $N_{f,w}, N_{f,w+1}, N_w, N_{w+1} \in \mathbb{N}$ with $N_w > 0$ and $N_{w+1} > 0$. If $N_{f,w} \cdot N_{w+1} \leq N_{f,w+1} \cdot N_w$, then as rationals,*

$$\frac{N_{f,w}}{N_w} \;\leq\; \frac{N_{f,w+1}}{N_{w+1}}.$$

Proof. Cross-multiplication: apply `div_le_div_iff` (mathlib) after casting to $\mathbb{Q}$, and use `exact_mod_cast` on the hypothesis. □ □

Theorem 4.3 (Step reduction for P_L; Lean: `monotone_step_of_PairCardIneq`)**.** *If $|\mathrm{wt}(w)| > 0$ and $|\mathrm{wt}(w+1)| > 0$ and $\mathrm{PairCardIneq}_{\mathcal{C},D}(w)$ holds, then $P_L^w(\mathcal{C}, D) \leq P_L^{w+1}(\mathcal{C}, D)$.*

Proof. Unfold P_L^w (which returns the ratio when the denominator is positive), then apply Lemma **??**. □ □

Theorem 4.4 (Chained monotonicity reduction; Lean: `monotone_of_PairCardIneq`)**.** *Suppose $w_1 \leq w_2$; suppose $|\mathrm{wt}(w)| > 0$ for all $w \in [w_1, w_2]$; and suppose $\mathrm{PairCardIneq}_{\mathcal{C},D}(w)$ holds for every $w \in [w_1, w_2)$. Then $P_L^{w_1}(\mathcal{C}, D) \leq P_L^{w_2}(\mathcal{C}, D)$.*

Proof. By induction on the Nat.le relation, using Theorem **??** at each step. □ □

This completes the abstract reduction. The entire question of monotonicity at weight $w \to w+1$ is now a purely integer inequality about two Finset cardinalities.

5 What Remains Open

We have reduced monotonicity to the pair-cardinality inequality (Definition **??**). Whether this inequality holds is still open in general.

5.1 It is not universally true

For abstract stabilizer codes, $\mathrm{PairCardIneq}_{\mathcal{C},D}(w)$ can fail at specific weights. Brute-force enumeration of small codes reveals:

- For the $[[5, 1, 3]]$ perfect code: $P_L^2 = 1 > P_L^3 = 7/9 > P_L^4 = 2/3$.
- For the Steane $[[7, 1, 3]]$ code: $P_L^2 = 7/9 > P_L^3 = 11/15$.
- For the rotated surface code at $d = 3$ (data-only, 9 qubits): small oscillations in the saturation tail ($P_L^5 \approx 0.7654 > P_L^6 \approx 0.7627$).

These counterexamples all occur either for perfect codes (where every weight-$(t+1)$ error fails) or in the saturation regime of small codes.

5.2 The target is circuit-level compiled codes

The physically relevant setting for our application is the *circuit-level compiled surface code*: the stabilizer code obtained from the full compiled memory-experiment circuit (data qubits, ancillae, r rounds of syndrome extraction, measurement errors, hook errors). This code has $C(d, r) \approx 8d^2 r$ fault locations, which for even small (d, r) places it well beyond brute-force enumeration.

The data-only counterexamples above do *not* apply at circuit level. Empirically (Stim Monte-Carlo experiments), circuit-level P_L^w is smooth and consistent with a rising S-curve followed by saturation, so PairCardIneq is expected to hold on the rising portion; but we have no proof.

The Lean library captures this cleanly: a file `Surface.lean` defines an abstract property `IsLatticeCode` (stabilizer weights bounded above, logical- error weights bounded below), and states a conjectural theorem that monotonicity holds on $[0, n - 2\ell_{\log}]$ for such codes. This conjecture is the one remaining `sorry` in the library.

5.3 Summary of open questions

1. Does $\mathrm{PairCardIneq}_{\mathcal{C},D}(w)$ hold for the circuit-level compiled surface code at $d \geq 3$ and all $w \in [0, n-2\ell_{\log}]$? This is the specialisation of the pair-cardinality inequality to the physically relevant setting.

2. Is there a clean structural hypothesis on $\mathcal{C}$ (e.g. CSS, translation-invariant, LDPC with local stabilizers) under which PairCardIneq is provable by a coupling or FKG-type argument?

3. A universal-approximation theorem for the three-parameter S-curve family within an appropriate regularity class of functions $[0, n] \to [0, 1]$ remains to be developed.

6 Conclusion

We have set up a minimal formal framework for reasoning about the weight-conditional logical error rate P_L^w of stabilizer codes under perfect MWPM. Four structural properties of P_L^w are unconditionally proved. The monotonicity question is reduced, cleanly and formally, to a single integer inequality on Finset cardinalities. The original aim — a first-principles derivation of the S-curve functional form — is not reached, but the present framework isolates the remaining combinatorial content and makes it amenable to either code-specific proof or machine-checked enumeration.

References

[1] J. Z. Ye and J. Palsberg, *ScaLER: Scalable Logical Error Rate Testing for Quantum Error Correction*, 2025.